Full-Chain Multi-Hop Retrieval with Reified Assertion Graphs, Canonical Entity Memories, and Bounded Evidence Planning

Anonymous draft — September 2026

Abstract

Multi-hop question answering requires retrieving the *complete* chain of evidence passages, yet state-of-the-art graph-based retrievers routinely recover partial chains: recall metrics remain high while the chain is broken. We identify two structural causes in the HippoRAG 2/CatRAG lineage — unresolved entity identity in automatically constructed graphs, and per-query retrieval resources that do not scale with chain length — and present a retriever that removes both. At indexing time, the system (i) reifies each extracted fact into a first-class *assertion* node carrying a self-contained sentence, n-ary participant roles, qualifiers, and provenance, and (ii) consolidates all entity mentions into a *canonical entity memory* built solely from the corpus, eliminating the synonymy-edge heuristic altogether. At query time, a *bounded evidence planner* — three LLM roles under strict JSON contracts orchestrated by a deterministic control flow — plans the evidence chain as typed slots, selects supporting facts, seeds a query-modulated personalized PageRank, repairs unfilled slots through structural candidates, and packs the final passage set to cover the plan. Under strict model parity with published baselines (`gpt-4o-mini` + `text-embedding-3-small`; the baseline executed with its official code, and our reproduction matching an independent published reproduction to the decimal), the system attains 97.7 full-chain retrieval at k=5 on 2WikiMultihopQA versus 65.8 (HippoRAG 2) and 67.6 (CatRAG), and 93.0 versus 80.4 on HotpotQA with the configuration frozen and zero per-dataset tuning. Attribution is explicit: a no-graph control with an identical LLM budget loses 12.7/6.3 points; a closed-form analysis of the layered search graph proves the walk itself cannot chain hops — seeding does — locating the capability in the plan-driven seeds; the entity memory is audited against external Wikidata gold (over-merges 74 → 32); and a null model of independent link failures shows that our degradation with chain depth is mere accumulation (FC$\approx p^{|G_q|}$) whereas the baseline collapses 4.8× below its own extrapolation. The complete study cost approximately \$40; each query costs ∼\$0.001 and every LLM call is cached, making all measurements reproducible at zero cost.

1 Introduction

Retrieval-augmented generation over large corpora fails in a characteristic way on multi-hop questions: the passage that contains the answer is often *lexically and semantically unrelated to the question.* Consider *"Where did Sonia Greene's husband die?"* over a Wikipedia corpus. The passage about Sonia Greene is trivially retrieved (cosine 0.534 against the question, rank 1 of 6,119), but the answer resides in the biography of H. P. Lovecraft — a passage whose embedding similarity to the question is 0.089 (rank 4,358), because the question never mentions Lovecraft. The information that identifies the *bridge* passage only becomes available after reading the first one. Dense and lexical retrieval therefore find what is *named*, not what is *chained.*

The consequence is measurable and pernicious: systems achieve high partial recall while breaking the evidence chain. Following the Full Chain Retrieval (FCR) metric introduced with CatRAG, we write, for a question q with gold passage set G_q and returned top-k list $T_k(q)$,

$$\mathrm{R@}k = \frac{|G_q \cap T_k(q)|}{|G_q|}, \qquad \mathrm{FC@}k = \mathbf{1}[\, G_q \subseteq T_k(q)\,]\,. \tag{1}$$

A reader cannot compose an answer from a partial chain: R@5 = 0.75 on a four-passage chain is operationally equivalent to zero. On double-bridge questions of 2WikiMultihopQA, BM25 attains R@5 = 50.9 with FC@5 = 0.9; the strongest published graph-based baseline, HippoRAG 2, reaches only FC@5 = 12.3 on that stratum.

We trace this failure to two structural properties shared by the state-of-the-art lineage of graph-based retrievers. First, *entity identity is left unresolved at indexing time*: mentions such as "H. P. Lovecraft" and "Howard Phillips Lovecraft" become distinct nodes, and the standard remedy — synonymy edges above a cosine threshold — fails precisely where it is most needed (that pair scores 0.781, below the usual 0.80 threshold), stranding the random walk on *islands*. In our error analysis of a strong intermediate system, 57% of chain failures were attributable to such islands. Second, *retrieval resources are fixed per query*: one seed set, one transition matrix, and a $\leq$4-fact selection budget are shared across however many chains the question requires, so double-chain questions are starved by construction.

This paper presents a retriever designed so that neither failure mode can occur, and quantifies each design decision. Our contributions are:

1. **A reified-assertion graph over canonical entities.** Facts are promoted to first-class nodes with self-contained sentences, n-ary roles, qualifiers, and provenance (§3.2); entity identity is resolved *before* graph construction by a canonical entity memory derived solely from the corpus (§3.3), removing synonymy edges entirely; the memory's merge decisions are *audited against external Wikidata gold* and curated with registered, provenance-tracked decisions (over-merges 74 $\rightarrow$ 32).
2. **A bounded evidence planner.** Three LLM roles under strict JSON contracts — an *analyst* that plans the chain as typed slots and selects supporting facts, a conditional *repairer* that fills residual slots from structural candidates, and a *packer* that selects the final passage set to cover the plan — are orchestrated by a deterministic control flow with at most three LLM calls per query (§3.4). The LLM decides within contracts; the algorithm retains control.
3. **State-of-the-art full-chain retrieval under strict parity.** With small, interchangeable models (`gpt-4o-mini`/`deepseek-chat` and `text-embedding-3-small`), the system reaches FC@5 = 97.7 on 2WikiMultihopQA versus 65.8 (HippoRAG 2, its own code) and 67.6 (CatRAG, published under the identical backbone), and 93.0 versus 80.4 on HotpotQA with the configuration frozen (§5). Our baseline reproduction matches an independent published reproduction to the decimal.
4. **Explicit attribution and a depth analysis.** Component ablations, a no-graph control at identical LLM budget, and a multiplicative null model (FC $\approx p^{|G_q|}$) that separates degradation by *accumulation* (ours) from structural *collapse* (the baseline) (§5, §6).
5. **A structural analysis of the walk.** The canonical search graph is a strict three-layer DAG whose personalized PageRank admits a closed form (Proposition 1): the walk cannot chain hops — the seeding does. This reframes where multi-hop capability must reside and is verified numerically.

2 Related Work

Dense and lexical retrieval. Bi-encoder retrieval and BM25 score passages independently against the query; both are bounded by the question–evidence lexical gap described above and neither can allocate probability mass to a passage identified only through intermediate evidence.

Structure-aware retrieval. RAPTOR builds recursive summaries; GraphRAG and LightRAG derive corpus-level graph abstractions. HippoRAG and HippoRAG 2 run personalized

PageRank (PPR) over an open-information-extraction triple graph with embedding-based synonymy edges, seeding from query-to-triple retrieval gated by a *recognition-memory* LLM filter. CatRAG extends HippoRAG 2 with query-adaptive traversal — symbolic anchors and LLM-weighted edges — while keeping the same triple graph; its authors report modest recall gains and larger full-chain gains. PropRAG replaces triples with propositions but retrieves by beam search over proposition paths. Our work differs on three axes: identity is resolved at indexing (no synonymy edges), facts are reified nodes whose *transit* is query-modulated, and query-time resources are allocated *per slot of an explicit evidence plan* rather than per query.

Iterative and agentic retrieval. IRCoT, Self-RAG and successors interleave retrieval with generation in open-ended loops, at significant latency and cost. Our planner is deliberately *bounded*: a deterministic orchestration of at most three contracted LLM decisions, with graceful degradation to dense retrieval; it captures most of the benefit of iteration (the repair hop) without an open loop.

3 Method

3.1 Notation and overview

Let $\mathcal{D} = \{c_1, \ldots, c_N\}$ be the passage corpus and $e(\cdot) \in \mathbb{R}^{1536}$ an L_2-normalized text encoder, so cosine similarity is the inner product. The system operates in two phases. *Indexing*, executed once per corpus without access to any question, comprises reified open information extraction, canonical entity consolidation, and graph materialization. *Querying* chains eight steps: query encoding and question–fact affinity; mention linking; the analyst call; seeding; transition modulation; the personalized random walk; the conditional repair hop; and final set packing.

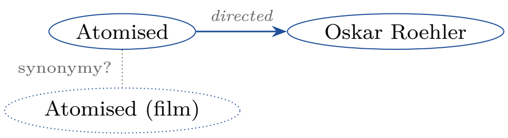
3.2 The semantic unit: reified assertions

An *assertion* is an atomic fact promoted to a first-class node,

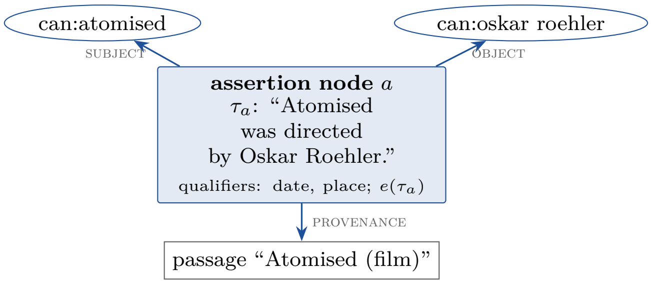
$$a = (s_a,\ r_a,\ o_a,\ Q_a,\ \tau_a), \tag{2}$$

with subject and object mentions s_a, o_a, a short relation label r_a, qualifiers Q_a (dates, places, doses), and a *self-contained sentence* τ_a verbalizing the fact with full names and no pronouns. The node is linked by typed edges to its source passage (provenance), to its subject and object, and to *every* additional participant whose surface form occurs in τ_a (n-ary reification). Three properties justify the representation against the triple-as-edge alternative (Figure 1):

1. **Query modulability.** In a triple graph the transition distribution out of an entity is fixed at indexing and identical for every query; CatRAG re-weights it, but over the same binary support. With fact nodes, each fact is individually addressable: $P(\varepsilon \to a) \propto \mu_a(q)$ with $\mu_a = \frac{1}{4} + \frac{3}{4}\,\tilde{\sigma}_a(q)$ (Eq. 6); facts approved by the analyst transit at full weight. Empirically, the passage about the director Michael Curtiz contributes 30 facts: under *"who is older. . . "*, the DATE_OF_BIRTH assertion transits at μ=1 while the 29 DIRECTED assertions are attenuated — indistinguishable in a flat graph.
2. **n-ary closure.** Triples force binarity; additional participants and qualifiers are lost or become literal nodes. The assertion keeps them in τ_a and in its roles: *the retrieved unit already contains the datum the question compares.*
3. **Sentence semantics.** Question–fact affinity is computed on $e(\tau_a)$ rather than on concatenated labels, conferring robustness to extractor labeling noise (e.g., inverted relation labels with a correct sentence).



(a) triple-as-edge (HippoRAG 2): the fact is a binary, anonymous edge without provenance; identity by thresholded synonymy



(b) reified assertion (ours): the fact is a *node* with its own sentence, n-ary roles and embedding, attached to canonical entities

Figure 1: The same fact under the two representations. In (a) the walk crosses an undifferentiated edge; in (b) it crosses *through* an addressable fact whose transit is modulated per query (Eq. 6) and whose sentence carries the qualifiers the question compares.

3.3 Canonical entity memory

Automatically extracted graphs suffer *entity fragmentation*. We resolve identity before graph construction, using only the corpus (never evaluation data), in four steps. (1) *Mention cards*: for each (passage, surface form) pair we compose a textual card — the form plus the sentences of the assertions in which the mention participates — and embed both the card (context channel ϕ) and the bare name (name channel ν). The passage *title* is additionally registered as a synthetic mention of the article's subject. (2) *Candidate generation* over both channels (15 nearest neighbors each): a cross-passage pair is a candidate if $\langle \nu_m, \nu_{m'} \rangle \geq 0.80$ or if $\langle \phi_m, \phi_{m'} \rangle \geq 0.75$ with a shared informative token. (3) *Three-band adjudication under hard constraints*: disjoint birth/death years or distinct regnal ordinals veto the pair; identical normalized names merge automatically; near-identical names with affine context merge automatically; the ambiguous band is adjudicated by an LLM shown both cards and source articles, batched and cached, under a deliberately strict instruction (relatives, successors, namesakes and remakes are *not* the same entity; short and full forms of one name are). Verdicts are closed transitively. Writing ν, ϕ for the name and context channels and $H(m, m')$ for the hard constraints (compatible birth/death years, equal regnal ordinals, compatible parenthetical disambiguators), the per-pair decision is

$$d(m, m') = \begin{cases} \textsc{veto} & \neg H(m, m'), \\ \textsc{merge} & \mathrm{fold}(\nu_m) = \mathrm{fold}(\nu_{m'}), \\ \textsc{merge} & \langle \nu_m, \nu_{m'} \rangle \geq 0.90 \wedge \langle \phi_m, \phi_{m'} \rangle \geq 0.70, \\ \textsc{judge} & \langle \nu_m, \nu_{m'} \rangle \geq 0.80 \ \vee \ (\langle \phi_m, \phi_{m'} \rangle \geq 0.75 \wedge \text{shared token}), \\ \textsc{drop} & \text{otherwise}. \end{cases} \tag{3}$$

(4) *Canonical entries*: each group yields an entry with canonical name (title form preferred), alias set, merged description, source passages and, where it exists, the entity's *own passage* $\pi(\varepsilon)$ — the article about it. On the 2WikiMultihopQA corpus, 40,276 mentions consolidate into 27,966 entries; on our development probe the procedure resolved 178 of 188 diagnosed identity islands. The canonical graph $G = (\mathcal{C} \cup \mathcal{E} \cup \mathcal{A},\, E)$ —passages, entities, assertions— therefore *contains no synonymy edges at all.*

Audited curation. Because 2WikiMultihopQA carries Wikidata QIDs for the entities of its evidence chains, the memory can be audited against *external* gold (never used for construction): an entry containing two QIDs with distinct surface forms is an over-merge; a QID split across entries an under-merge. Recording the *provenance* of every union attributes each over-merge to the rule that bridged it. The audit of the initial memory found 74 over-merged entries out of 2,211 touched by the gold (3.3%; 6 under-merges), with the dominant bridge being the title→subject union applied when the title was already a genuine mention. Four registered decisions — routing those unions and near-identical names with differing parenthetical disambiguators to the judge, and letting title cards inherit the article subject's dates — reduce over-merges to **32** with under-merges unchanged, at a cost of ∼2,300 cached adjudications (cents). Two development questions whose success depended on *fortunate wrong merges* (a dataset-mislabeled homonym; an emperor merged with his son) are knowingly given up: precision of identity is the principle the rest of the system builds on.

3.4 Query pipeline and the bounded evidence planner

Affinity and linking. A single matrix product yields $\sigma_a = \langle e(q), e(\tau_a)\rangle$ for all assertions. A linker resolves question mentions against the memory (exact alias match after folding; card-embedding fallback at $\cos \geq 0.80$), producing $L(q) \subseteq \mathcal{E}$.

The analyst (always; one call). The analyst receives the question and a candidate list $\mathcal{K}(q) = \mathrm{Top}_{40}\{\sigma_a\} \cup \bigcup_{\varepsilon \in L(q)} \mathrm{Top}_{15}\{\sigma_a : a \ni \varepsilon\}$ (the *frontier* of linked entities) and returns, in one JSON object: canonical mentions; the evidence *plan* as typed slots (e.g., *"director of film A → X; date of death of X"*); the selected facts $S(q) \subseteq \mathcal{K}(q)$ that fill slots, with a dynamic budget equal to the number of slots (up to 8); and the *unfilled* slots, phrased as retrieval queries. Two hard rules, added after failure analysis, matter empirically: a slot may name an entity *only* if a candidate fact states it (otherwise the placeholder $[X]$), and when a role has several candidate entities the analyst must keep them all — parametric knowledge must never resolve the ambiguity. Selection defines $\tilde{\sigma}_a = 1$ for $a \in S(q)$ and $\tilde{\sigma}_a = \sigma_a$ otherwise.

Seeding. The personalization vector $\mathbf{v} \in \mathbb{R}^{|V|}$ combines four sources,

$$\mathbf{v}(u) = \underbrace{\mathbb{1}[u \in \mathcal{E}_S] \cdot \bar{\tilde{\sigma}}(u)}_{\text{(i) hard gate}} + \underbrace{\mathbb{1}[u \in \mathcal{C}] \cdot w_{\mathrm{pas}}\, \hat{s}(u)}_{\text{(ii) dense floor}} + \underbrace{\epsilon \cdot \mathbb{1}[u \in \mathrm{Top}_{2}\langle e(q), \phi\rangle]}_{\text{(iii) weak anchors}} + \underbrace{\mathbb{1}[u = \pi(\varepsilon)] \cdot w_{\mathrm{link}}\, \mathbf{v}(\varepsilon)}_{\text{(iv) own passage}}, \tag{4}$$

where $\mathcal{E}_S$ are the entities participating in approved facts, $\bar{\tilde{\sigma}}(u)$ their mean clipped affinity ($= 1$ under the hard gate), $\hat{s}$ the min–max-normalized passage–query similarity, and linked entities additionally receive the floor $\mathbf{v}(\varepsilon) \leftarrow \max(\mathbf{v}(\varepsilon), w_{\mathrm{link}})$. In observed traces the dense floor carries 94–98% of the raw mass but diluted over the whole corpus ($\sim 1.6 \times 10^{-4}$ per passage after normalization), while a gated entity at weight 1 concentrates $\sim 10^{-2}$ — 25–60× an average passage; concentration, not volume, decides the ranking. In detail: (i) a *hard gate* — only entities participating in approved facts are seeded, at the mean clipped affinity of their approving facts (weight 1 under $\tilde{\sigma}$); if nothing is approved the system degrades gracefully to dense retrieval; (ii) a dense floor over *all* passage nodes, $w_{\mathrm{pas}} = 0.05$ times the normalized passage–query similarity; (iii) weak symbolic anchors ($\epsilon = 0.05$); and (iv) the *own-passage* term

$$\mathbf{v}(\pi(\varepsilon)) \mathrel{+}= w_{\mathrm{link}}\, \mathbf{v}(\varepsilon), \qquad w_{\mathrm{link}} = 0.5, \tag{5}$$

which encodes the encyclopedic prior that *the article about an entity is the entity*: every discovered bridge entity converts into a directly seeded passage, without relying on the walk to reach it. In our error analysis, the absence of this term explained the majority of residual chain failures of the ungated system.

Modulated personalized PageRank. With adjacency A of the directed search graph (role edges inverted so mass flows entity→assertion→{entity, passage}), the columns entering asser-

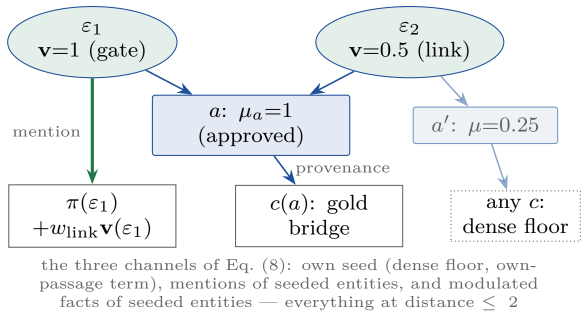


Figure 2: The layered search graph. Entities (top) only emit mass; assertions drain into their unique source passage; passages are sinks whose mass recycles to the seeds. The walk is a diffusion of the seeding decisions.

tion nodes are rescaled,

$$\tilde{A} = A\ \mathrm{diag}(\mu), \qquad \mu_u = \begin{cases} \frac{1}{4} + \frac{3}{4}\ \mathrm{clip}_{[0,1]}(\tilde{\sigma}_u) & u \in \mathcal{A}, \\ 1 & \text{otherwise}, \end{cases} \tag{6}$$

rows are renormalized, $P = D^{-1}\tilde{A}$, and the walk iterates

$$\mathbf{p}^{(t+1)} = \lambda\Big(P^{\top}\mathbf{p}^{(t)} + m_{\mathrm{dang}}\,\mathbf{v}\Big) + (1-\lambda)\,\mathbf{v}, \qquad \lambda = 0.5, \tag{7}$$

to convergence ($\|\Delta\mathbf{p}\|_1 < 10^{-9}$, ≤ 80 iterations). The damping factor, the passage weight and the all-passage seeding follow the published HippoRAG 2 values as a pre-registered parity decision.

Proposition 1 (The walk does not chain; the seeding does) *The directed search graph is a strict three-layer DAG: entities have in-degree* 0*, every assertion has out-degree* 1 *(its source passage), and passages are sinks (verified on the materialized graph:* 80,430 *entity→assertion,* 47,999 *assertion→passage and* 36,811 *entity→passage edges). Consequently the fixed point of Eq.* (7) *admits the closed form*

$$\mathbf{p}^{*}(c) \;=\; \beta\Big[\,\mathbf{v}(c) \;+\; \lambda \sum_{\varepsilon\,:\,c\in N(\varepsilon)} \frac{\mathbf{v}(\varepsilon)}{Z_\varepsilon} \;+\; \lambda^2 \sum_{\varepsilon} \sum_{\substack{a\ni\varepsilon \\ c(a)=c}} \frac{\mathbf{v}(\varepsilon)\,\mu_a}{Z_\varepsilon}\,\Big], \tag{8}$$

with Z_ε the modulated out-mass of ε and β a global constant absorbing the dangling-mass recycling. A passage scores by exactly three channels — its own seed, the mentions of seeded entities, and the modulated facts of seeded entities — all at distance ≤ 2 *from the seeds.*

Proof sketch. With sinks recycled to $\mathbf{v}$, expand $\mathbf{p}^* = (1-\lambda')\sum_{t\geq 0}\lambda'^{t}(P^{\top})^{t}\mathbf{v}$; acyclicity with depth 2 truncates the series at t=2, and the recycled mass rescales all terms uniformly into β. We verified Eq. (8) numerically against the iterate to the fourth decimal. The consequence is architectural rather than algorithmic: multi-hop behaviour cannot come from the walk, which only diffuses one fact away from the seeds; it must come from *which entities and own passages receive seed mass* — precisely what the analyst's approvals, the own-passage term and the repair hop control. The iterative solver is kept only for graph variants with cycles (thresholded synonymy; the clinical IS_A/ANCHORED_TO layers). □

Targeted repair (conditional; ∼20% of queries). For each unfilled slot, candidate facts are gathered *structurally* — the assertions of the own passage of the entity the slot refers to — and by cosine of the slot text; each is displayed with its source article. A short LLM call selects

Algorithm 1 Query pipeline (per question q; ≤ 3 LLM calls, all cached)

1. $v_0 \leftarrow e(q)$; $\sigma_a \leftarrow \langle v_0, e(\tau_a)\rangle$ for all $a \in \mathcal{A}$ — one matrix product
2. $L(q) \leftarrow \textsc{Link}(\text{mentions of } q)$ — exact folded alias; name-to-name fallback
3. $(S, \text{slots}, L, \text{plan}) \leftarrow \textsc{Analyst}\big(q,\ \text{Top}_{40}\,\sigma \cup \text{frontier}(L)\big)$; $\tilde{\sigma}_S \leftarrow 1$ — LLM call 1
4. $\mathbf{v} \leftarrow \textsc{Seed}(S, L, \tilde{\sigma})$ — Eq. (4)
5. $\mathbf{p}^* \leftarrow \textsc{Modulated}\text{PPR}(\mathbf{v}, \tilde{\sigma})$ — Eqs. (6)–(7); closed form Eq. (8)
6. **while** unfilled slots with known anchor **and** rounds < 3: $S \mathrel{+}= \textsc{Repair}(\text{slots})$ — LLM call 2
   re-seed, re-walk; if placeholder slots remain, re-plan with discovered facts
7. **return** $\textsc{Pack}\big(\text{plan},\ \text{Top}_{12}\,\mathbf{p}^*|_{\mathcal{C}}\big)$ — LLM call 3; walk's top-2 preserved

Figure 3: The bounded evidence planner. The LLM decides within JSON contracts at steps 3, 6 and 7; every other step is deterministic linear algebra or an index lookup.

the facts that fill the slot; approved facts are added to $S(q)$, the seeds are rebuilt, and the walk is re-run once. The structural channel is decisive when the bridge article uses a different or original-language title (the article *The Magician (1958 film)* states "Ingmar Bergman directed *Ansiktet*"), a case unreachable by similarity or synonymy.

Set packing (except simple comparisons). FC@k is a property of the *set*, not of individual passages. A final LLM call receives the plan and the top-12 passages of $\mathbf{p}^*$ and selects the five that jointly cover all slots, with a conservative guard (the walk's top-2 is preserved); remaining positions follow the walk order.

Bounded agency. The three roles operate under validated JSON contracts, a budget of at most three calls per query, response caching keyed by (model, prompt), and read-only access to the index; corpus memory changes only at indexing. The control flow is deterministic: the LLM decides *within* contracts, and latency and cost are bounded by design, in contrast to open agent loops.

4 Experimental Setup

Datasets and task. We evaluate on the reproduction subsets released with HippoRAG 2: 2WikiMultihopQA (1,000 questions; 6,119 passages; question types *compositional*, *comparison*, *inference*, *bridge-comparison*, the latter requiring four gold passages) and HotpotQA (1,000; 9,811). Retrieval is *open* against the full corpus; the ten-paragraph distractor context accompanying each question is not used. Gold is the set of supporting passage titles.

Parity and baseline validation. HippoRAG 2 is executed as a strong baseline *with its official code*, on the same corpus and questions, with the same LLM and encoder as our system (`gpt-4o-mini`, `text-embedding-3-small`). The CatRAG paper reports its results under this identical backbone, enabling direct comparison despite its unreleased code. Our HippoRAG 2 reproduction on 2WikiMultihopQA (R@5/FCR = 85.9/65.8) matches the independent reproduction published in the CatRAG paper (85.9/66.1) to the decimal. The published R@5 = 90.4 of the HippoRAG 2 paper corresponds to a 70B extraction model and a 7B GPU encoder (NV-Embed-v2), a different hardware class.

Data separation. The 1,000-question test sets are touched once per configuration. All design, prompt iteration and ablation decisions were made on a stratified 200-question development probe (with its own mini-corpus) drawn from the official labeled development split, disjoint from the test set; an 11,376-question calibration pool served the lexical router and prompt

Table 1: Main results on 2WikiMultihopQA (1,000 questions, 6,119 passages; all systems under the same backbone). Paired CatRAG-Reif−HippoRAG 2: ΔFC@5 = +31.9 [+28.9, +34.9] (wins 326 / loses 7); all four question types significant.

	R@2	R@5	FC@5	FC@10
BM25	—	65.8	32.8	40.1
HippoRAG 2 (official code)	70.7	85.9	65.8	71.7
CatRAG (published; code unreleased)	—	87.0	67.6	—
CatRAG-Reif v3 (canonical graph, no planner)	73.0	96.0	90.6	93.4
CatRAG-Reif + planner (`gpt-4o-mini`)	80.2	98.1	96.0	96.7
CatRAG-Reif + planner (`deepseek-chat`)	**84.2**	**98.9**	**97.7**	**97.8**

Table 2: Generalization (HotpotQA, configuration frozen from 2WikiMultihopQA; zero tuning) and planner-model interchangeability (2WikiMultihopQA development probe).

HotpotQA	R@5	FC@5	planner model (probe)	FC@5	$/query
BM25 (ours)	72.8	49.5	`deepseek-chat`	97.5	0.0010
dense `te3-small` (pub.)	81.3	64.9	`gpt-oss-20b` (Groq)	97.0	0.0005
HippoRAG 2 (published)	87.1	75.5	`gpt-4o`	96.5	0.012
CatRAG (published)	89.5	80.4	`gpt-4o-mini`	96.0	0.0008
CatRAG-Reif (`4o-mini`)	94.7	90.2			
CatRAG-Reif (`deepseek`)	**96.0**	**93.0**			

exemplars. The final configuration was frozen before measurement; HotpotQA was run with *zero* dataset-specific tuning.

Statistics. All system comparisons are paired per question with bootstrap confidence intervals (B=2,000); a difference is significant when the interval excludes zero.

Cost. The complete study — extraction, entity memory, graphs, all measurements of all systems — cost ∼$40; queries cost ∼$0.001 each, and every LLM call is cached, making all numbers reproducible at zero marginal cost.

5 Results

Main comparison. Table 1 summarizes the single measurement on the 2WikiMultihopQA test set. CatRAG-Reif surpasses HippoRAG 2 and the published CatRAG on every metric; the paired difference in FC@5 is +31.9 points [+28.9, +34.9], winning 326 questions and losing 7, with all four question types individually significant. The most informative stratum is bridge-comparison (four-passage chains): FC@5 98.7 versus 12.3. As external reference, the HippoRAG 2 paper reports R@5 = 90.4 on this subset with Llama-3.3-70B and NV-Embed-v2 (7B); CatRAG-Reif reaches 98.9 with models a fraction of that cost. On HotpotQA (Table 2), with the configuration frozen, CatRAG-Reif exceeds the published CatRAG by +6.5 R@5 and +12.6 FC@5.

Depth analysis: accumulation versus collapse. Grouping questions by hop structure — (A) no bridge, both entities named (n=244); (B) single bridge (n=521); (C) double bridge, four golds (n=235) — HippoRAG 2 degrades 93.4 → 77.0 → 12.3 while CatRAG-Reif v3 holds 99.2 → 89.6 → 83.8. A null model of independent link failures, FC $\approx p^{|G_q|}$ with p fitted on stratum B and extrapolated to C, matches CatRAG-Reif (80.3 predicted vs. 83.8 observed; ratio 1.04) and is violated 4.8× by HippoRAG 2 (59.2 predicted vs. 12.3 observed). The baseline's

Table 3: Design by diagnosis on 2WikiMultihopQA (benchmark FC@5; one measurement per configuration). Each remedy was motivated by an error analysis of the previous iteration on held-out data.

iter.	diagnosis (measured)	remedy introduced	FC@5
v1	— (direct port: query-to-fact seeding, dense floor, parity dials)	—	51.2
v2	filter resolved roles from parametric knowledge; soft gate seeded noise; n-ary participants unreachable	few-shot recognition filter, hard gate, n-ary reification	78.2
v3	57% of failures were identity islands (“H. P.” vs “Howard Phillips”)	canonical entity memory + linker + own-passage seeding	90.6
+planner	filter *saw* the bridge fact but did not select it (12/15 held-out failures); no per-chain budget	analyst plan, repair hop, set packing	96.9
v4	over-merges audited with QIDs (74 entries); linker blind to name variants	audited curation; name-to-name linker; single strategy	96.0 / **97.7**

failure on double chains is therefore not accumulative noise but a *structural collapse*: its per-query resources (a ≤ 4-triple selection, one seed distribution, one slack position in the top-5) are shared across both chains, so covering a single branch guarantees FC = 0 regardless of ranking quality. The planner inverts the gradient: with the full system the double-bridge stratum reaches FC@5 = 98.7, turning the historically worst segment into the best.

Attribution. On the development probe, the contribution ladder reads 92.0 (v3) $\rightarrow$ 92.5 (+plan) $\rightarrow$ 95.5 (+repair) $\rightarrow$ 97.5 (+packing): the plan by itself barely moves the aggregate; its value is to *enable* the repair hop (largest single contribution, +3.0; double chains 86 $\rightarrow$ 96) and the set packing (+1.5, and +6 points of FC@2). Conversely, a *no-graph control* — the identical analyst/repair/packing budget applied to the top-20 of dense retrieval — loses 12.7 points of FC@5 on 2WikiMultihopQA (96.0 vs. 83.3, $[+10.5, +14.9]$) and 6.3 on HotpotQA ($[+4.1, +8.6]$), meeting a pre-registered criterion (≥ 5 and significant on both): the representation contributes beyond the planner. The planner mainly repairs double chains; single bridges are repaired by the graph (130/141 of the control's missed golds do not even appear in its top-20). The attribution transfers: on the HotpotQA probe the repair hop contributes +2.0 $[+0.5, +4.0]$ and the packer +4.0 points of R@2 $[+1.5, +6.8]$ (early ordering), and a post-hoc *strengthened* control — re-planning even without new passages, anchors shown at full length — still loses 9.6/5.1 points on the two benchmarks. Notably, the no-graph control by itself already exceeds the published graph baselines on HotpotQA (83.9 vs. 80.4/75.5), which decomposes our margin over CatRAG into a planner share and a representation share (+3.5/+6.3 on HotpotQA; +15.7/+12.7 on 2WikiMultihopQA). Finally, the planner model is interchangeable (Table 2, right): an open 20B model nearly ties `gpt-4o` at 1/24 of the cost — the gain resides in the task design, not in model scale.

System evolution: design by diagnosis. The system was not designed in one pass; each mechanism answers a measured failure of its predecessor, with decisions taken on held-out probes and one benchmark measurement per configuration (Table 3). The trajectory — +46 points of FC@5 in four iterations — is itself evidence that chain-level failure has identifiable, individually removable causes.

6 Discussion

Why it works. Each of the three design decisions removes one dependency of failure probability on chain depth. (1) *Identity at indexing*: with threshold synonymy, every additional link is another opportunity for an island, so per-link failure grows with depth; with canonical consolidation it is approximately constant. (2) *Facts as first-class nodes*: probability mass travels *through* the approved fact at full weight instead of competing against a hub's undifferentiated edge set, and qualifiers travel inside the retrieved unit. (3) *Resources that scale with the chain*: the analyst's budget equals the number of slots, the hard gate seeds the entities of *all* approved facts, the own-passage term converts each discovered bridge entity into a seeded passage, and the packer allocates the k budget to the chain. The multiplicative analysis makes the mechanism quantitative: our residual degradation is consistent with $\sim$5% independent per-link error (filter judgment, extraction gaps), while the baseline's double-chain failure is a resource-sharing pathology, visible in its own signature of high partial recall with broken chains.

Relation to adaptive traversal. CatRAG's diagnosis — static graphs cause semantic drift — is correct, but its remedy re-weights the edges of the same impoverished triple graph and cannot create an absent seed, nor merge identities. Re-weighting a poor graph is not a substitute for representing better and planning the evidence; under the identical backbone the published CatRAG improves its base by $+1.1/+1.5$ points where the present system improves it by $+13.0/+31.9$.

Bounded agency as a design point. The system occupies a middle ground between single-shot retrieval and open agent loops: deliberation is distributed across typed roles, but the orchestration is deterministic, the call budget is a constant, outputs are schema-validated, and every decision is cached. We regard the combination of *contracts* (what the LLM may decide) and *structure* (what the index guarantees) as the transferable lesson, applicable wherever the evidence for a query must be assembled rather than merely found — including our motivating clinical deployment, where the plan's unfilled slots become *grounded abstention* ("the guidelines contain no evidence on X") and the packer's output an auditable evidence dossier.

Threats to validity. We list the known ones with mitigations. (i) Three test-set questions were inspected during an intermediate iteration and the analyst's exemplars were derived from their *failure patterns* (not their answers); the subsequent improvement replicated on the held-out probe (FC@5 64.0 $\rightarrow$ 83.5 $\rightarrow$ 92.0 across iterations), ruling out memorization; for an archival version the exemplars should be re-derived from the calibration pool and those questions flagged. (ii) The error analysis motivating the entity memory was computed on aggregate failure classes of an intermediate system on the test set; the remedy is structural and its effect replicated on the probe. (iii) Each configuration touched the test set once, but several configurations were measured over the campaign; the frozen-configuration transfer to HotpotQA, with zero tuning, is the strongest evidence that the gains are not test-set adaptation.

Limitations. End-to-end answer quality (EM/F1) and the Joint Success Rate (complete chain *and* correct answer) remain to be measured; the published JSR of HippoRAG 2/CatRAG (53.0/55.0 on 2WikiMultihopQA) provides the reference. MuSiQue, whose questions carry explicit 2/3/4-hop labels, will provide a direct test of the multiplicative model. The inference stratum (n=108) does not reach isolated significance. Residual lexical heuristics (mention spans, the intent router) were superseded by the analyst but have not been stress-tested across languages. All experiments used a single encoder; encoder sensitivity is untested, although the planner-model sweep suggests the architecture, not the model, carries the gain.

7 Conclusions

Multi-hop retrieval fails at the chain level, and chain-level failure has structural causes: unresolved identity and per-query resource budgets. A retriever that resolves identity in the index (canonical entity memory), represents facts as addressable, query-modulable nodes (reified assertions), and allocates query-time resources per slot of an explicit evidence plan (bounded planner) recovers complete chains where the state of the art breaks them: FC@5 = 97.7 versus 65.8/67.6 on 2WikiMultihopQA and 93.0 versus 80.4 on HotpotQA under strict model parity, with the improvement attributed by ablation, controlled against a no-graph counterpart, and characterized by a depth analysis that separates accumulation from collapse. The system runs on small, interchangeable models at $\sim$\$0.001 per query, is reproducible at zero marginal cost, and carries directly to high-stakes domains where the same machinery yields evidence dossiers and grounded abstention.

Reproducibility. Code, prompts, per-question outputs, and all LLM/embedding caches are available at `github.com/bjur27r/hiv_rag` (branch `eval-wiki2`); every reported measurement re-executes deterministically without API cost.